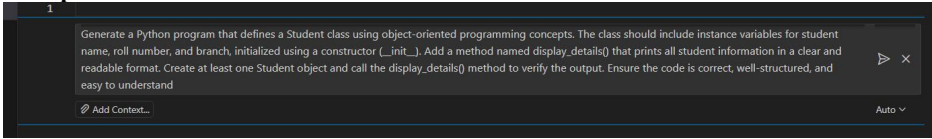


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| AssignmentNumber:6.3(Present assignment number)/24(Total number of assignments) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                           |
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| Q.No.                                                                           | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Expected Time to complete |
| 1                                                                               | <p><b>Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals</b></p> <p><b>Lab Objectives</b></p> <ul style="list-style-type: none"><li>• To explore AI-powered auto-completion features for core Python constructs such as classes, loops, and conditional statements.</li><li>• To analyze how AI tools suggest logic for object-oriented programming and control structures.</li><li>• To evaluate the correctness, readability, and completeness of AI-generated Python code.</li></ul> <p><b>Lab Outcomes (LOs)</b></p> <p>After completing this lab, students will be able to:</p> <ul style="list-style-type: none"><li>• Use AI tools to generate and complete Python class definitions and methods.</li><li>• Understand and assess AI-suggested loop constructs for iterative tasks.</li><li>• Generate and evaluate conditional statements using AI-driven prompts.</li><li>• Critically analyze AI-assisted code for correctness, clarity, and efficiency.</li></ul> <hr/> <p><b>Task Description #1: Classes (Student Class)</b></p> <p><b>Scenario</b></p> <p>You are developing a simple student information management module.</p> <p><b>Task</b></p> <ul style="list-style-type: none"><li>• Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class.</li><li>• The class should include attributes such as name, roll number, and branch.</li><li>• Add a method display_details() to print student information.</li><li>• Execute the code and verify the output.</li><li>• Analyze the code generated by the AI tool for correctness and clarity.</li></ul> <p><b>Expected Output #1</b></p> <ul style="list-style-type: none"><li>• A Python class with a constructor (__init__) and a display_details() method.</li><li>• Sample object creation and output displayed on the console.</li><li>• Brief analysis of AI-generated code.</li></ul> <p><b>Prompt:</b></p>  <p><b>Code:</b></p> | Week3 - Wednesday         |

```
AI assignment_63.py X
AI assignment_63.py > ...
1 class Student:
2     """A class to represent a student."""
3
4     def __init__(self, name, roll_number, branch):
5         """
6         Initialize student with name, roll number, and branch.
7
8         Args:
9             name (str): Student's name
10            roll_number (int): Student's roll number
11            branch (str): Student's branch of study
12        """
13        self.name = name
14        self.roll_number = roll_number
15        self.branch = branch
16
17    def display_details(self):
18        """Display all student information in a readable format."""
19        print("-" * 40)
20        print("Student Information")
21        print("-" * 40)
22        print(f"Name: {self.name}")
23        print(f"Roll Number: {self.roll_number}")
24        print(f"Branch: {self.branch}")
25        print("-" * 40)
26
27    # Create Student objects and display their details
28    student1 = Student("Alice Johnson", 101, "Computer Science")
29    student1.display_details()
30
31    student2 = Student("Bob Smith", 102, "Electronics")
32    student2.display_details()
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\abbua\OneDrive\Desktop\AI ASSISTED CODING> & c:\Users\abbua\AppData\Local\Microsoft\WindowsApps\python3.11.exe "c:\Users\abbua\OneDrive\Desktop\AI ASSISTED CODING\AI assignment_63.py"
-----
Student Information
-----
Name: Alice Johnson
Roll Number: 101
Branch: Computer Science
-----
Student Information
-----
Name: Bob Smith
Roll Number: 102
Branch: Electronics
-----
PS C:\Users\abbua\OneDrive\Desktop\AI ASSISTED CODING>
```

### Explanation:

This code defines a Student class used to store and display basic student information. The class has a constructor (`__init__`) that initializes three attributes: name, roll number, and branch when a student object is created. The `display_details()` method neatly prints the student's information with clear labels and separators, making the output easy to read. At the end of the program, two Student objects are created with different details, and the `display_details()` method is called for each object to show their information on the console. Overall, the code is simple, well-structured, and clearly demonstrates how classes and objects work in Python.

### Task Description #2: Loops (Multiples of a Number)

#### Scenario

You are writing a utility function to display multiples of a given number.

#### Task

- Prompt the AI tool to generate a function that prints the first 10 multiples of a given number using a loop.
- Analyze the generated loop logic.
- Ask the AI to generate the same functionality using another controlled looping structure (e.g., while instead of for).

#### Expected Output #2

- Correct loop-based Python implementation.
- Output showing the first 10 multiples of a number.
- Comparison and analysis of different looping approaches.

#### Prompt:

Generate a Python function that accepts an integer as input and prints the first 10 multiples of that number using a for loop. Ensure the loop logic is correct and the output is properly formatted. Explain how the loop iterates to generate the multiples. Then generate another Python function that performs the same task using a while loop instead of a for loop. Finally, compare both looping approaches in terms of readability, control, and suitability for this task.

Add Context...

Auto

## Code:

```
AI assignment_6.3.py > print_multiples_for
1 def print_multiples_for(n):
2     """Prints the first 10 multiples of the given integer using a for loop."""
3     for i in range(1, 11):
4         print(f"{n} x {i} = {n * i}")
5
6 def print_multiples_while(n):
7     """Prints the first 10 multiples of the given integer using a while loop."""
8     i = 1
9     while i <= 10:
10        print(f"{n} x {i} = {n * i}")
11        i += 1
12
13 # Example usage
14 number = int(input("Enter an integer: "))
15 print_multiples_for(number)
16 print_multiples_while(number)
17
18
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + - - - - -

```
PS C:\Users\abbua\OneDrive\Desktop\AI ASSISTED CODING> & C:/Users/abbua/AppData/Local/Microsoft/WindowsApps/python3.11.exe "c:/Users/abbua/OneDrive/Desktop/AI ASSISTED CODING/ai_assignment_6.3.py"
Enter an integer: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
PS C:\Users\abbua\OneDrive\Desktop\AI ASSISTED CODING>
```

## Explanation:

This code defines two functions to print the first 10 multiples of a given number using different looping structures. The `print_multiples_for()` function uses a for loop that runs from 1 to 10 and prints the multiplication result for each value. The `print_multiples_while()` function performs the same task using a while loop, where a counter variable starts at 1 and increments until it reaches 10. The program then asks the user to enter an integer and calls both functions, showing how the same logic can be implemented using both for and while loops in Python.

## Task Description #3: Conditional Statements (Age Classification)

### Scenario

You are building a basic classification system based on age.

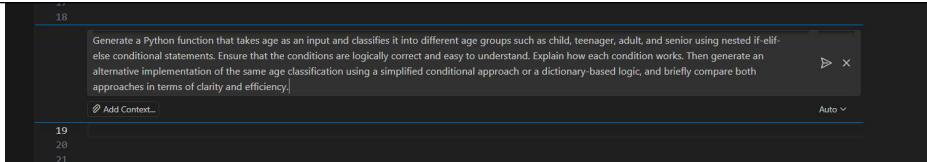
### Task

- Ask the AI tool to generate nested if-elif-else conditional statements to classify age groups (e.g., child, teenager, adult, senior).
- Analyze the generated conditions and logic.
- Ask the AI to generate the same classification using alternative conditional structures (e.g., simplified conditions or dictionary-based logic).

### Expected Output #3

- A Python function that classifies age into appropriate groups.
- Clear and correct conditional logic.
- Explanation of how the conditions work.

### Prompt:



## Code:

```
[1] ✓ On  # Approach 1: Nested if-elif-else conditional statements
def classify_age_nested(age):
    """
    Classifies age into groups using nested if-elif-else statements.
    - Child: 0-12 years
    - Teenager: 13-19 years
    - Adult: 20-64 years
    - Senior: 65+ years
    """
    if age < 0:
        return "Invalid age"
    elif age <= 12:
        return "Child"
    elif age <= 19:
        return "Teenager"
    elif age <= 64:
        return "Adult"
    else:
        return "Senior"

# Approach 2: Dictionary-based logic
def classify_age_dict(age):
    """
    Classifies age using a dictionary with tuples as keys.
    More concise and easier to modify age ranges.
    """
    age_groups = [
        (0, 12, "Child"),
        (13, 19, "Teenager"),
        (20, 64, "Adult"),
        (65, 150, "Senior")
    ]
```

```
[1] ✓ On  ]

for min_age, max_age, group in age_groups:
    if min_age <= age <= max_age:
        return group
return "Invalid age"

# Example usage
if __name__ == "__main__":
    test_ages = [5, 15, 30, 70, -1]

    print("\nNested if-elif-else approach:")
    for age in test_ages:
        print(f"Age {age}: {classify_age_nested(age)}")

    print("\nDictionary-based approach:")
    for age in test_ages:
        print(f"Age {age}: {classify_age_dict(age)}")

... Nested if-elif-else approach:
Age 5: Child
Age 15: Teenager
Age 30: Adult
Age 70: Senior
Age -1: Invalid age

Dictionary-based approach:
Age 5: Child
Age 15: Teenager
Age 30: Adult
Age 70: Senior
Age -1: Invalid age
```

## Explanation:

This code shows two different ways to classify a person's age into groups such as child, teenager, adult, and senior. In the first approach, the `classify_age_nested()` function uses if-elif-else statements to check the age step by step and return the appropriate category, while also handling invalid negative ages. In the second approach, the `classify_age_dict()` function stores age ranges and their labels in a list of tuples and uses a loop to find where the given age fits, making the logic more compact and easier to modify. In the example usage section, a list of test ages is checked using both functions, and the results are printed to compare how each approach works.

## Task Description #4: For and While Loops (Sum of First n Numbers)

### Scenario

You need to calculate the sum of the first n natural numbers.

### Task

- Use AI assistance to generate a `sum_to_n()` function using a for loop.

- Analyze the generated code.
- Ask the AI to suggest an alternative implementation using a while loop or a mathematical formula.

Expected Output #4

- Python function to compute the sum of first n numbers.
- Correct output for sample inputs.
- Explanation and comparison of different approaches.

Prompt:

```
71
Generate a Python function named sum_to_n(n) that calculates and returns the sum of the first n natural numbers using a for loop. Ensure the code is correct, readable, and handles valid inputs properly. Explain how the loop works to compute the sum. Then generate an alternative implementation of the same function using a while loop or a mathematical formula, and compare the different approaches in terms of simplicity, efficiency, and readability.
```

Code:

```
C:\> Users > abbu > aa.py > ...
1 # Approach 1: Using a for loop
2 def sum_to_n(n):
3     """Calculate sum of first n natural numbers using a for loop."""
4     total = 0
5     for i in range(1, n + 1):
6         total += i
7     return total
8
9 # Test cases
10 print(sum_to_n(5)) # Output: 15
11 print(sum_to_n(10)) # Output: 55
12 print(sum_to_n(100)) # Output: 5050
13
14
15 # Approach 2: Using a while loop
16 def sum_to_n_while(n):
17     """Calculate sum of first n natural numbers using a while loop."""
18     total = 0
19     i = 1
20     while i <= n:
21         total += i
22         i += 1
23     return total
24
25 print(sum_to_n_while(5)) # Output: 15
26
27
28 # Approach 3: Using mathematical formula
29 def sum_to_n_formula(n):
30     """Calculate sum of first n natural numbers using formula: n(n+1)/2."""
31     return n * (n + 1) // 2
32
33 print(sum_to_n_formula(5)) # Output: 15
34
35
36 # Comparison
37 if __name__ == "__main__":
38     # All three return the same result
39     assert sum_to_n(100) == sum_to_n_while(100) == sum_to_n_formula(100)
40     print("All approaches verified!")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\abbu> & C:\Users\abbu\AppData\Local\Microsoft\WindowsApps\python3.11.exe c:\Users\abbu\aa.py

15  
55  
5050  
15  
All approaches verified!  
All approaches verified!  
PS C:\Users\abbu>

Explanation:

This code shows **three different ways** to calculate the sum of the first  $n$  natural numbers. The first function uses a **for loop**, where numbers from 1 to  $n$  are added one by one to a total variable. The second function uses a **while loop**, which repeats the addition process until the counter reaches  $n$ . The third function uses a **mathematical formula**  $n(n+1)/2$ , which gives the result instantly without looping, making it the fastest method. Test cases check whether the functions give correct outputs, and the final assertion confirms that all three approaches produce the same result, proving their correctness.

Task Description #5: Classes (Bank Account Class)

Scenario

You are designing a basic banking application.

Task

- Use AI tools to generate a Bank Account class with methods such as deposit(), withdraw(), and check\_balance().

- Analyze the AI-generated class structure and logic.
- Add meaningful comments and explain the working of the code.

Expected Output #5

- Complete Python Bank Account class.
- Demonstration of deposit and withdrawal operations with updated balance.
- Well-commented code with a clear explanation.

Prompt:

```
Generate a Python BankAccount class for a basic banking application. The class should include an __init__ constructor to set account holder name and initial balance, and methods deposit(amount), withdraw(amount) (with balance checking), and check_balance(). Add clear comments explaining each part of the code. Provide example usage that demonstrates deposits, withdrawals, and updated balance outputs. After generating the code, include a brief explanation of how the class works and evaluate the structure, logic, readability, and efficiency of the AI-generated implementation
```

Code:

```
84 # BankAccount class for basic banking operations
85 class BankAccount:
86     """A simple banking account class to manage deposits and withdrawals."""
87
88     def __init__(self, account_holder, initial_balance=0):
89         """
90         Initialize a bank account with account holder name and initial balance.
91
92         Args:
93             account_holder (str): Name of the account holder
94             initial_balance (float): Starting balance (default: 0)
95         """
96         self.account_holder = account_holder
97         self.balance = initial_balance
98
99     def deposit(self, amount):
100         """
101         Deposit money into the account.
102
103         Args:
104             amount (float): Amount to deposit (must be positive)
105         """
106         if amount > 0:
107             self.balance += amount
108             print(f"Deposited ${amount:.2f}. New balance: ${self.balance:.2f}")
109         else:
110             print("Deposit amount must be positive.")
111
112     def withdraw(self, amount):
113         """
114         Withdraw money from the account with balance verification.
115
116         Args:
117             amount (float): Amount to withdraw
118         """
119         if amount > self.balance:
120             print(f"Insufficient funds. Available balance: ${self.balance:.2f}")
121
122         elif amount > 0:
123             self.balance -= amount
124             print(f"Withdrew ${amount:.2f}. New balance: ${self.balance:.2f}")
125         else:
126             print("Withdrawal amount must be positive.")
127
128     def check_balance(self):
129         """
130         Display current account balance.
131         """
132         print(f"Account holder: {self.account_holder}")
133         print(f"Current balance: ${self.balance:.2f}")
134
135 # Example usage
136 account = BankAccount("Alice", 500)
137 account.check_balance()
138 account.deposit(200)
139 account.withdraw(100)
140 account.withdraw(1000) # Insufficient funds
141 account.check_balance()
```

```
78
79
80 if amount > self.balance:
81     print(f"Insufficient funds. Available balance: ${self.balance:.2f}")
82 elif amount > 0:
83     self.balance -= amount
84     print(f"Withdrew ${amount:.2f}. New balance: ${self.balance:.2f}")
85 else:
86     print("Withdrawal amount must be positive.")
87
88 def check_balance(self):
89     """Display current account balance."""
90     print(f"Account holder: {self.account_holder}")
91     print(f"Current balance: ${self.balance:.2f}")
92
93 # Example usage
94 account = BankAccount("Alice", 500)
95 account.check_balance()
96 account.deposit(200)
97 account.withdraw(100)
98 account.withdraw(1000) # Insufficient funds
99 account.check_balance()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
All approaches verified!
Account holder: Alice
Current balance: $500.00
Deposited $200.00, New balance: $700.00
Withdrew $100.00, New balance: $600.00
Insufficient funds. Available balance: $600.00
Insufficient funds. Available balance: $600.00
Account holder: Alice
Current balance: $600.00
PS C:\Users\abbas>
```

Explanation:

This code defines a **BankAccount** class that simulates basic banking operations like depositing, withdrawing, and checking balance. The `__init__` method initializes the account holder's name and starting balance. The `deposit()` method adds money to the account only if the amount is positive, while the `withdraw()` method subtracts money after checking if there are enough funds and if the amount is valid. The `check_balance()` method displays the account holder's name and current balance. In the example usage, an account is created for "Alice" with an initial balance of 500, then deposits and withdrawals are performed, including a failed withdrawal due to insufficient funds, showing how the class manages transactions and updates the balance.